Unified Engineering Workbench

Pass Science Works R&D Internship Submission

Candidate ID: _____ 2 November 2026

SUMMARY

Offline C++17 prototype with separate thermal, import, performance, hardware mock and backpropagation modules. All computation, parsing and concurrency are native C++17. Results export under results/.

PUBLISHING STATUS

No public URL is invented in this local report. Before submission, upload the repository and Release ZIP to GitHub, replace this note with verified public links, test them in a private browser, and regenerate the PDF.

GEOMETRY AND RESULT IMPORT

ASCII STL facets and exact CSV header node_id,x_m,y_m,z_m,temperature_C are validated. Empty, incomplete and non-finite input is rejected. Tested limits are 1,000,000 STL lines and 1,000,000 nodes. Supplied samples are self-created; the CSV is synthetic, not an ANSYS export. Current limit: the console interface does not provide interactive 3D rotation/pan/zoom/rendering.

THERMAL FEA VERIFICATION

Uniform insulated rod: L=1 m, A=1e-4 m2, k=10 W/(m K), q=10000 W/m3, T(0)=100 C, T(L)=20 C. Two-node elements use kA/h [1 -1; -1 1] and qAh/2 [1 1]. Both prescribed endpoints are correctly eliminated. Exact: T=100-80x/L+q*x*(L-x)/(2k).

Elements	Nodes	Maximum midpoint error (C)	Ratio
10	11	1.25	-
20	21	0.3125	4.00
40	41	0.078125	4.00
80	81	0.01953125	4.00

Errors use FE midpoint interpolation, not only nodes. Raw curves and convergence values are in results/thermal_N*.csv and thermal_convergence.csv. Debug insight: nodal error is near round-off whereas linear midpoint interpolation of the quadratic field has a nonzero error that converges 4x.

PARALLEL PERFORMANCE

Seed 20261102; 20,000 independent cases; 80 elements; warm-up plus five timed repetitions. Machine reported 16 logical CPUs; workers capped at 4. Medians: 1=54.581 ms; 2=28.682 ms, 1.903x, 95.1%; 3=20.047 ms, 2.723x, 90.8%; 4=17.540 ms, 3.112x, 77.8%. Difference=0 C.

EMBEDDED DESIGN AND BACKPROPAGATION

Hardware behaviour is SIMULATED: MCU, FPGA and JTAG adapter discovery, capabilities, connect/disconnect and telemetry. FPGA demonstrates recoverable timeout. JTAG is a debug transport, not a generic board. Verify pinout, common ground, 3.3 V domain, reference voltage and level shifting physically.

For x=1, target=2: h=tanh(wx+b), yhat=v*h+c, J=0.5(yhat-y)^2. Initial (w,b,v,c)=(0.5,0,1,0), loss=1.18254. Analytical gradients are (-1.20946, -1.20946, -0.710682, -1.53788). Central finite differences with epsilon=1e-6 have max absolute error 4.61e-11. A simultaneous 0.1 learning-rate update is used; 100 loss samples are exported.

TEST EVIDENCE AND LIMITATIONS

Launch.exe test passed thermal boundary/convergence and valid/malformed non-finite input tests. Launch.exe all 200 40 completed all modules. Limitations: console-only UI, no interactive 3D plots, no physical hardware, and no verified public upload. Record final physical cores, RAM and launch-to-ready time honestly on the submission machine.

OWNERSHIP

Source and sample data are self-created. Standard CMake/MinGW libraries are used. AI assistance and a deliberate NaN-rejection challenge test are disclosed in docs/AI_USE_LOG.md. Attach handwritten chain-rule appendix.